Transformer u(t) Experiment Report

Manuscript time-dependent control trained by PMP/KKT optimality gaps

This report reproduces the manuscript's time-dependent Transformer control strategy u_theta(t): [0,T] -> [0,u_max].

1. Model and Training Objective

We use the population dynamics and cost from the manuscript:

$$\dot{N}_i(t) = \big(r_i - \phi_i u(t) - M_i G(N(t))\big) N_i(t) \tag{1}$$

$$G(N) = \log\left(1 + \frac{1}{m}\sum_{k=1}^{m} N_k\right) \tag{2}$$

$$J(u) = \alpha^\top N(T) + \int_0^T \big(\beta^\top N(t) + \gamma u(t)\big)\, dt \tag{3}$$

Reported parameters: T=10, m=21, u_max=3, alpha=1, beta=0.1, gamma=20, and N_i(0)=10. The control is represented by a small Transformer encoder over normalized time.

Given u_theta(t), we roll out N_theta(t), solve the costate equation backward, and compute the switching function

$$\psi(t) = H_u(N, \lambda, u) = \gamma - \sum_i \phi_i \lambda_i(t) N_i(t) \tag{4}$$

component	role
non-singular KKT loss	enforces u=0 when psi>0 and u=u_max when psi<0
singular loss	near psi=0, matches u_theta(t) to the singular candidate u_sing(N(t))

Here q(t) is a smooth weight that switches between the singular condition near psi(t)=0 and the boundary KKT condition away from psi(t)=0.

2. Learned u(t), N(t), and N(u)

The learned control starts high, transitions to a lower interior/singular-like region, and increases again near the terminal portion.

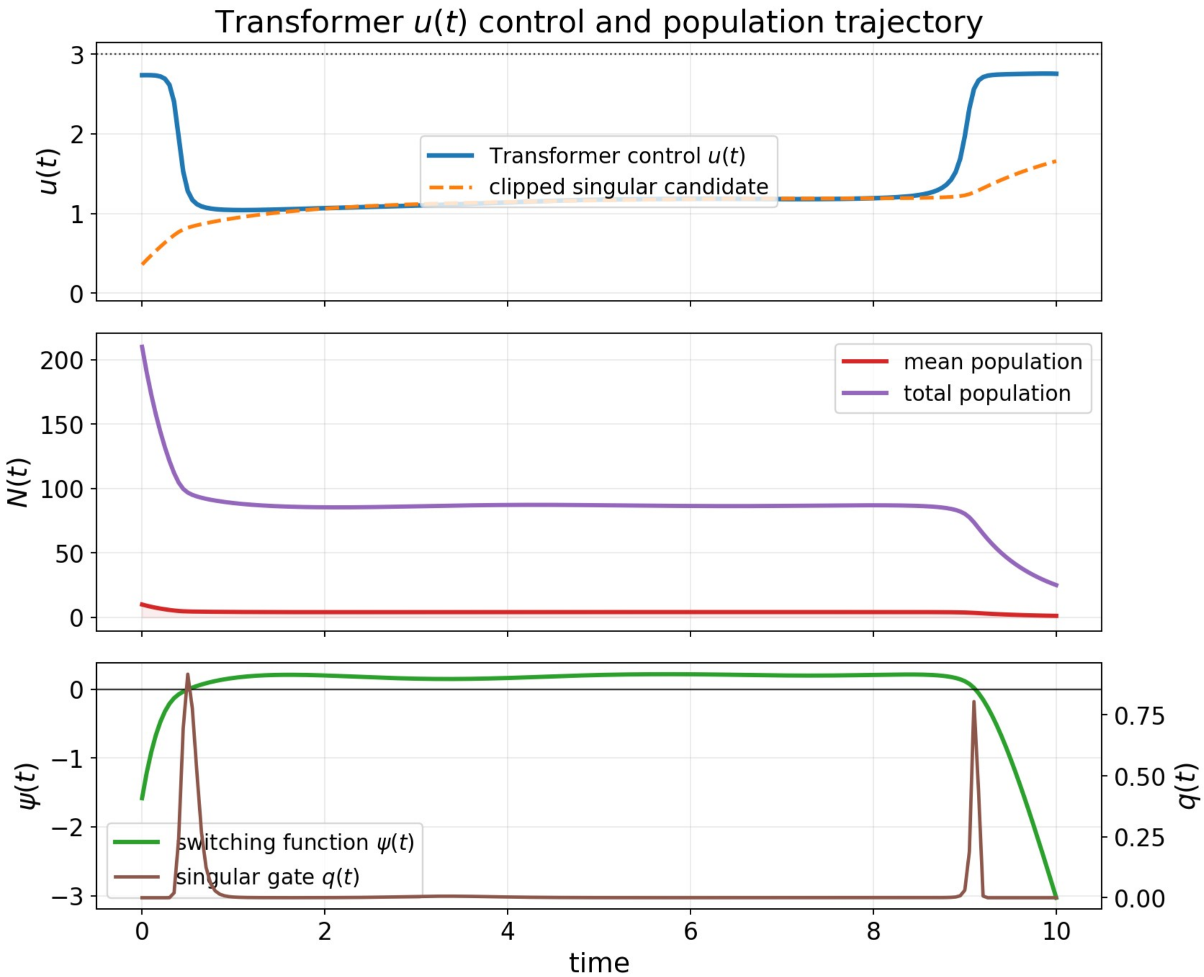

Figure 1. Learned Transformer control u(t), population trajectory N(t), switching function psi(t), and singular weight q(t).

The N(u) phase plot below uses the total population across all 21 subpopulations.

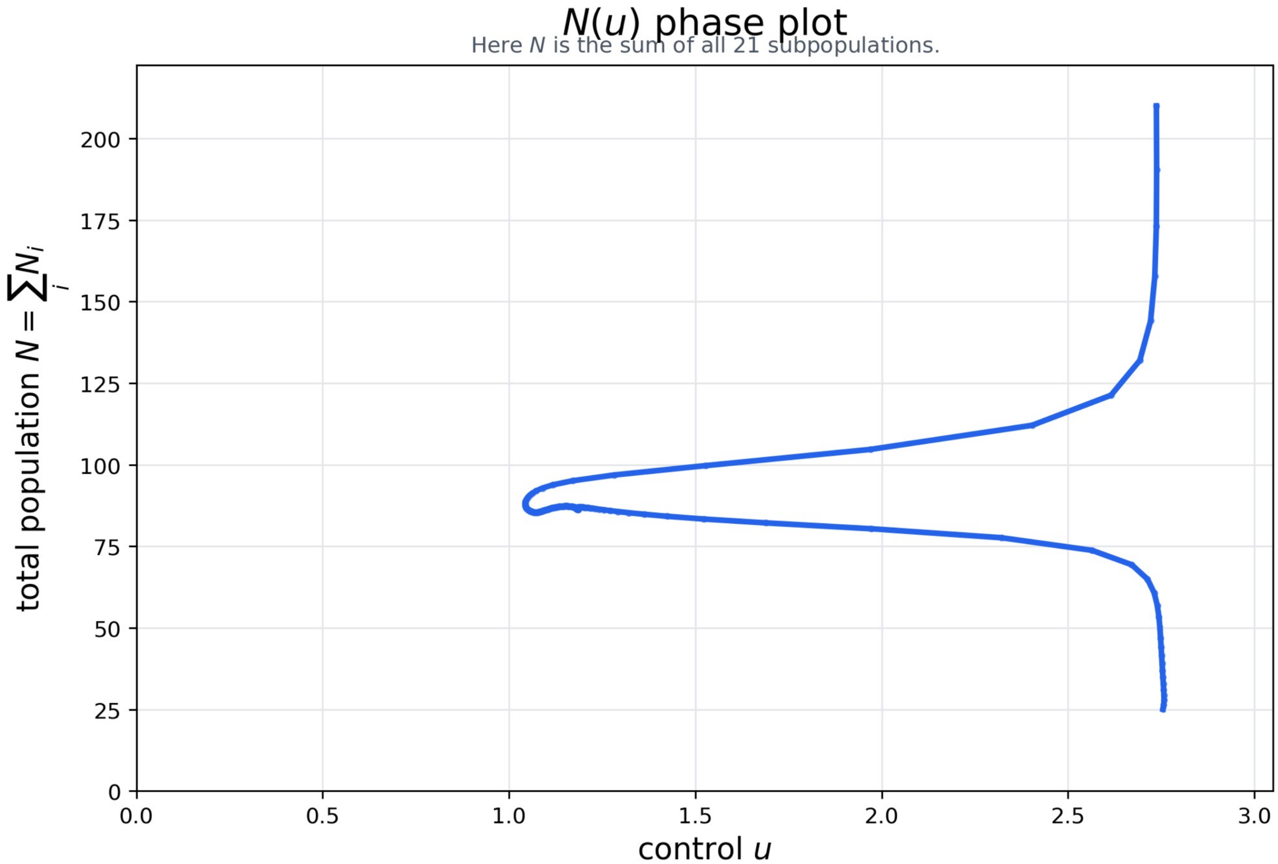

Figure 2. N(u) phase plot using the total population across the 21 subpopulations.

3. Training Loss Trajectories for PMP/KKT Conditions

The table reports the smallest recorded training loss. In this experiment, the training loss is the manuscript's PMP/KKT optimality gap, composed of the singular condition and the non-singular Hamiltonian minimization condition.

metric	value
total training loss / PMP-KKT optimality gap	0.02637
singular-condition training loss	0.00167
non-singular Hamiltonian-minimization training loss	0.02471
objective J on the training grid	384.76
range and mean of u_theta(t)	min 1.038, max 2.758, mean 1.372
terminal mean state	1.207

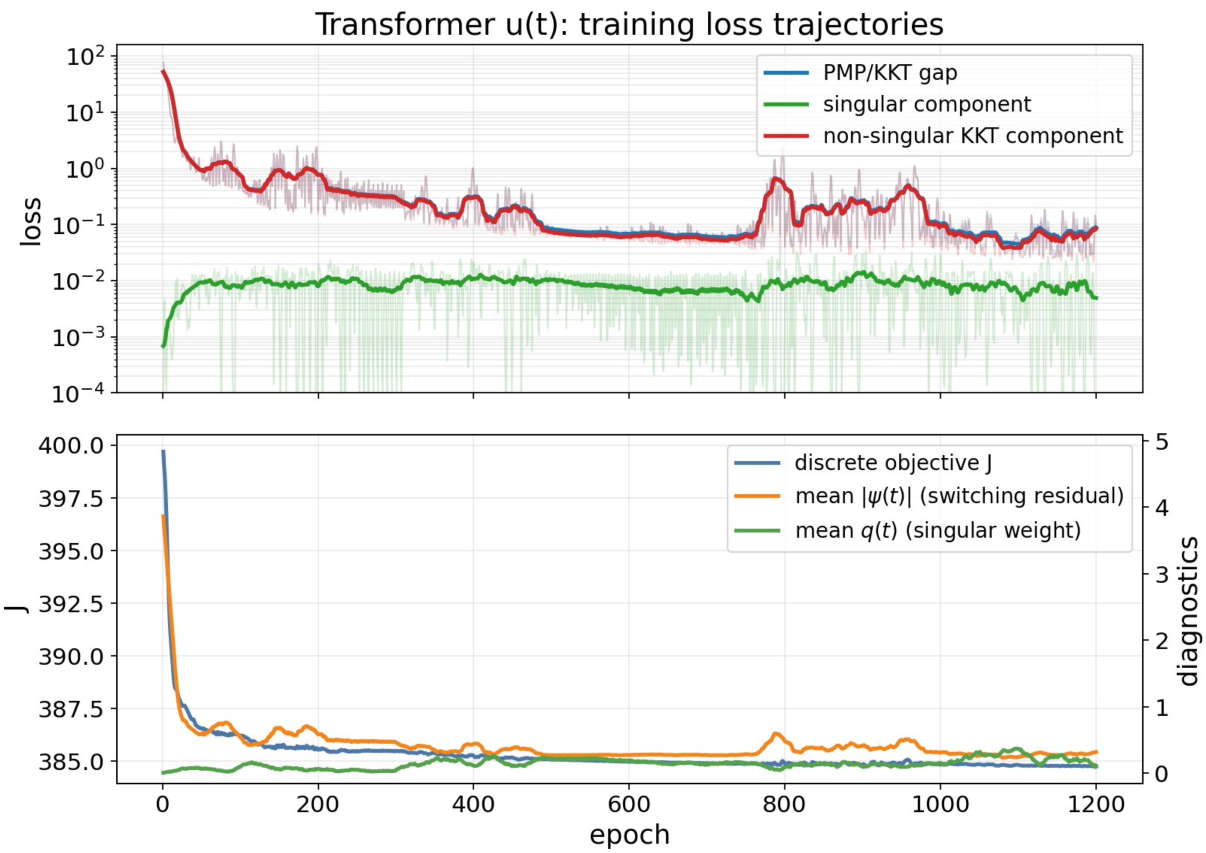

Figure 3. Training trajectories of the total PMP/KKT optimality gap and its singular and non-singular components.

The final pointwise diagnostic below uses a smooth weight q(t) to separate the two regimes: near psi(t)=0 it emphasizes the singular-condition error; away from psi(t)=0 it emphasizes the boundary KKT error.

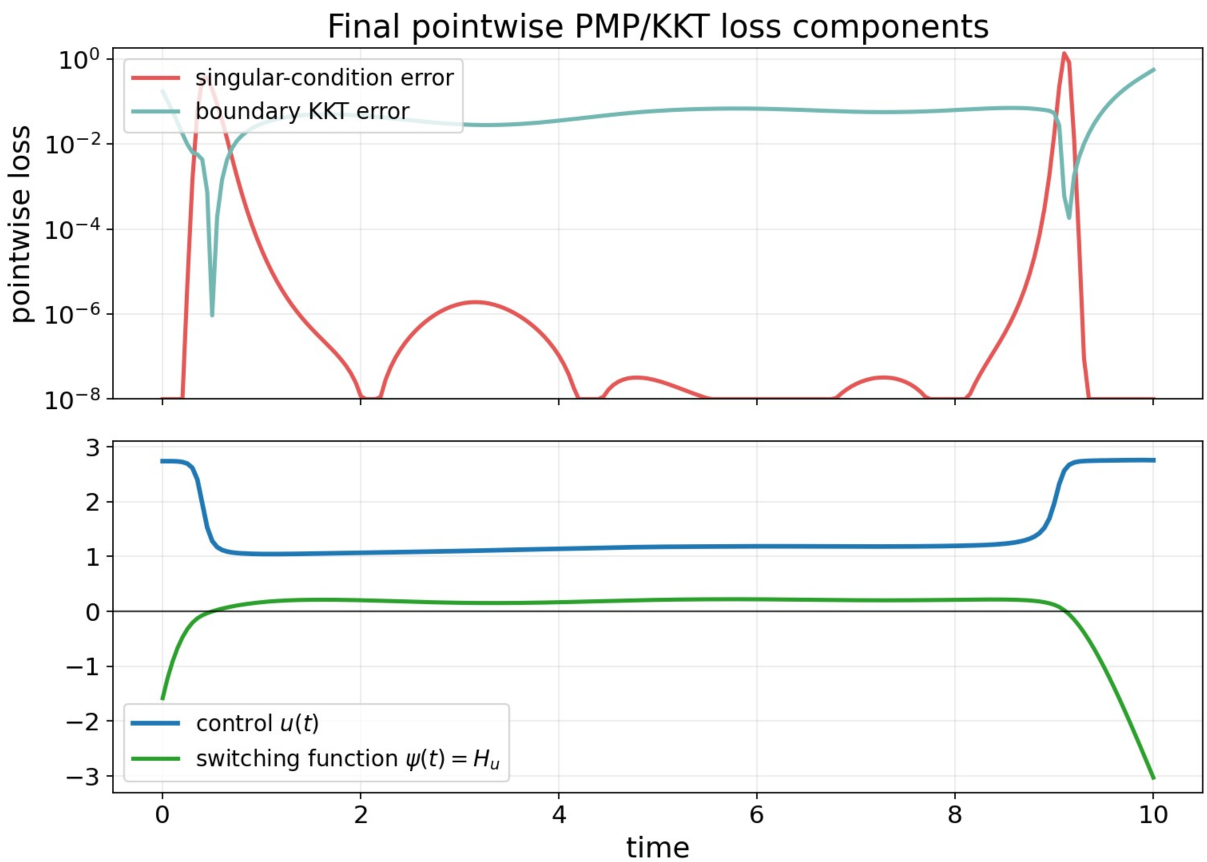

Figure 4. Pointwise singular-condition error and boundary KKT error along the final learned trajectory.

4. Comparison With Gu et al. [3]

Here [3] refers to Gu, Xiong, and Chen, Pontryagin Optimal Control via Neural Networks (arXiv:2212.14566). Their Neural-PMP method contains two parts: learning a differentiable dynamics model from data, and then using a PMP-gradient update to optimize the control sequence. Since the dynamics are known in our manuscript setting, we compare against an oracle-dynamics version of the second part: forward state integration, backward costate recursion, and Hamiltonian-gradient updates of a discrete control sequence. This row should therefore be read as a controller-stage baseline following [3].

method	control update / training criterion	PMP/KKT gap	objective J*	note
Transformer u(t)	paper PMP/KKT optimality-gap loss	0.0523	386.70	main u(t) reproduction
Neural-PMP controller-stage [3]	Hamiltonian-gradient update with known dynamics	7.47	387.02	related-work comparison
direct grid cost	minimize discretized J	0.805	386.47	reference only
constant u=1.5	fixed control	76.89	400.40	scale check

* The J values in this table are computed after fixing u(t), reintegrating N(t) with a finer-step fourth-order Runge-Kutta method, and then applying the manuscript objective definition. This is only to use the same numerical integration accuracy across methods.

Under the same PMP/KKT gap calculation, the Transformer u_theta(t) has a much smaller gap than this Neural-PMP controller-stage baseline. When J is recomputed with the same numerical evaluator, the Transformer also has a slightly lower objective value in this run.

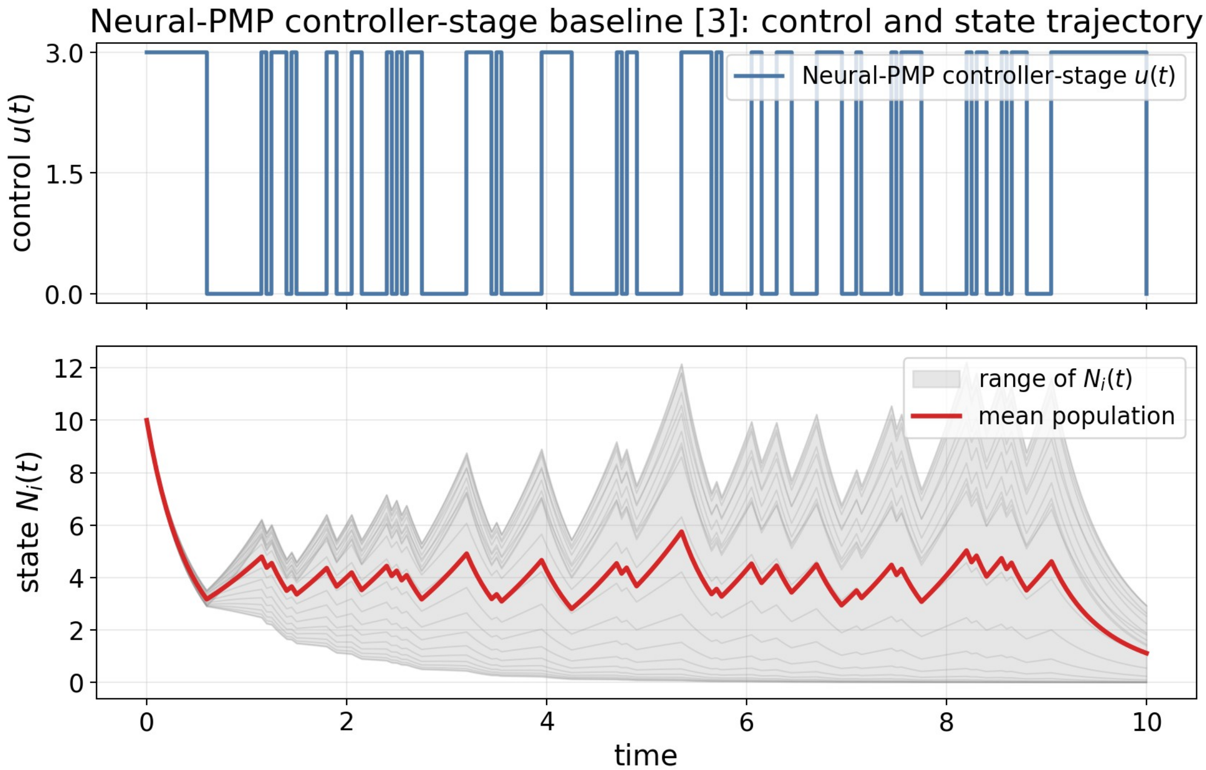

Figure 5. Neural-PMP controller-stage baseline [3]: control sequence and resulting state trajectory.

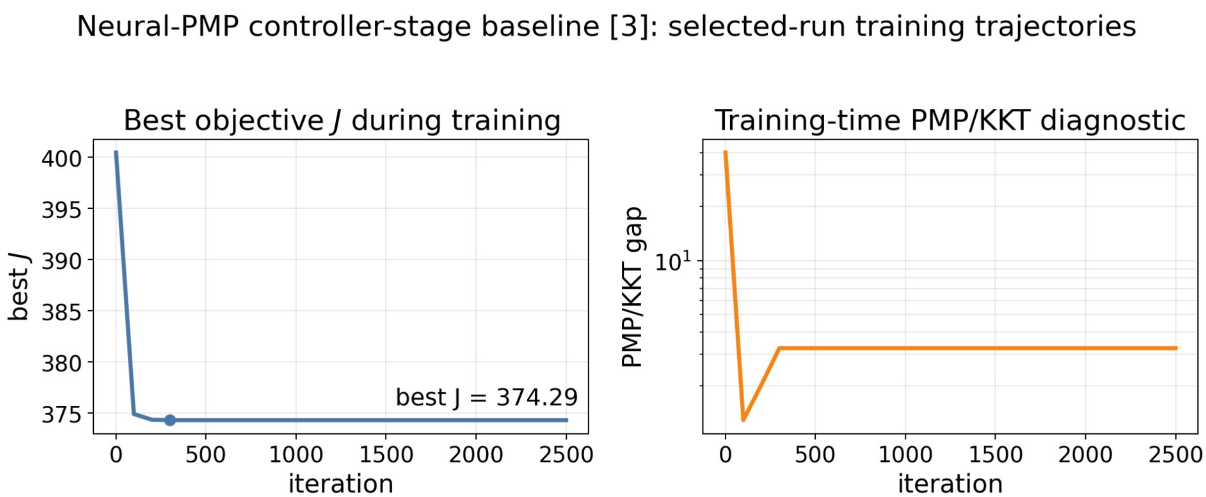

Figure 6. Neural-PMP controller-stage baseline [3]: selected-run training trajectories.

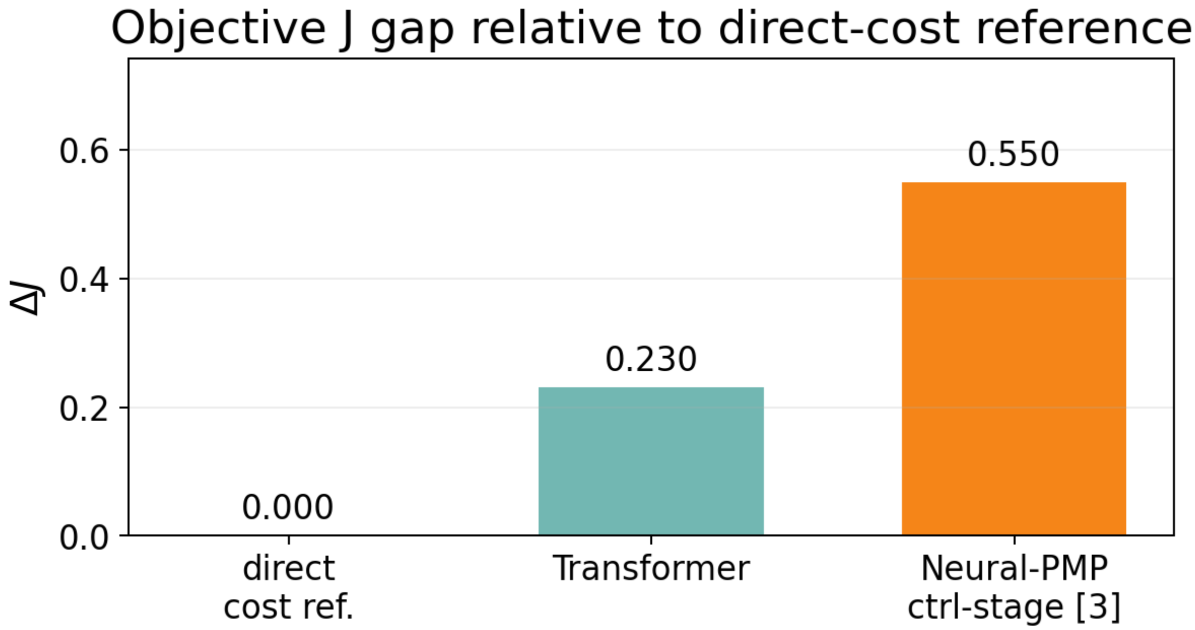

Figure 7. Objective J gap relative to the direct-cost reference under the common numerical evaluation.

5. Conclusion

The requested u(t) reproduction is complete. The Transformer control trajectory trained with the manuscript's PMP/KKT optimality-gap loss is smooth and satisfies the control bounds, reduces the training optimality gap from 76.26 to 0.02637, and gives J approximately 386.70 under the common numerical evaluation. The state-dependent extension u(t,N) is left for separate work because it requires different optimality conditions.